LIVE EVENT DETECTION FOR PEOPLE’S SAFETY USING NLP AND DEEP LEARNING

A SEMINAR REPORT

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SOLAMAN AJI
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MAR BASELIOS INSTITUTE OF TECHNOLOGY & SCIENCE
NELLIMATTOM, KERALA, INDIA -686693

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MAR BASELIOS INSTITUTE OF TECHNOLOGY AND SCIENCE NELLIMATTOM

CERTIFICATE

This is to certify that the seminar report entitled **“LIVE EVENT DETECTION FOR PEOPLE’S SAFETY USING NLP AND DEEP LEARNING”** Submitted by **SOLAMAN AJI** with Reg no: **MBI21CS056** during the academic year 2024-2025 towards the partial fulfillment of the requirement for the award of Bachelor of Technology in Computer Science & Engineering of APJ Abdul Kalam Technological University Trivandrum, Kerala is a bonafide work carried out by him in this department under our guidance and supervision.

Mrs.Nisreen M Ali	Mrs.Mintu Thomas	Mr. Midhun Mathew
Asst. Prof. CSE Dept	Asst. Prof. CSE Dept	Head of Department
Seminar Guide	Seminar Coordinator	Computer Science & Engineering

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ABSTRACT

In the physical world, the occurrence of any physical event would bear with it a sound, particular to that event only. It will be good to develop a system able to detect ambient noise and judge whether it is related to something positive or something negative. Today, humans pose the greatest threat to society by getting involved in robbery, assault, or homicide activities .Any such kind of threat in real time is always associated with a noise which may be used for an early detection. Numerous existing measures are available but none of them sounds efficient due to lack of accuracy, delays in exact prediction of threats. The solution for this problem can be attained with the help of sound based NLP(Natural Language Processing) techniques, which detects threats from person's surrounding sound, then an automatic and immediate alert with the geographical location should be shared to the registered contact or emergency services. The live recording module continuously listen for any sound from person's surroundings, and send the recorded sound to the prediction module. The prediction module analyses the sound is danger or not. If it is danger, automatic alert message is sent to the emergency services through WhatsApp, SMS, email,etc. else discards. To carry out accurate prediction, training and testing of Kaggle dataset is carried out using 3 deep learning models namely LSTM(Long Short-Term Memory), 1D CNN(1-Dimensional Convolution Neural Network), 2D CNN(2-Dimensional Convolution Neural Network) can achieve with a average accuracy of 96.03%.This can solve greatest practical problems humankind faces in today's world while exploring and delivering on the domain of deep learning and audio analysis for the detection of live events from ambient sounds.

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ABBREVATIONS

CNN	Convolution Neural Network
LSTM	Long Short-Term Memory
NLP	Natural Language Processing
ARM	Advanced RISC Machine
EDA	Exploratory Data Analytics
FFT	Fast Fourier Transform
STFT	Short Time Fourier Transform
GPU	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

Today, humans pose the greatest threat to society by getting involved in robbery, assault, or homicide activities. Such circumstances threaten the people working alone at night in remote areas especially women. Any such kind of threat in real time is always associated with a sound/noise which may be used for an early detection. A system capable of detecting and categorizing these sounds could help identify the nature of ambient noise in real-time. While such applications are not yet commonplace, the technology to analyse sounds and text through natural language processing is already available and used in various fields. Numerous existing measures are available but none of them sounds efficient due to lack of accuracy, delays in exact prediction of threat.

The solution of the problem utilizes sound-based NLP techniques, incorporating a live recording module and a prediction module to analyse ambient sounds. The live recording module continuously captures sounds from a person's surroundings and transmits them to the prediction module, which classifies the sounds using deep learning models—LSTM, 1D CNN, and 2D CNN—trained on a Kaggle dataset for accurate predictions. By analysing sound patterns, this system can differentiate between common sounds and those indicating potential danger, allowing for a proactive approach to safety. A significant benefit of this system is its reliance on a smartphone, eliminating the need for additional external hardware or wearable devices, making it convenient for users who only need to carry their phones. If the prediction module identifies a dangerous sound, such as a scream or crash, the system immediately and automatically sends an alert to emergency services, ensuring a rapid response to emergencies.

CHAPTER 2

LITERATURE SURVEY

2.1 BACKGROUND AND RELATED WORK

Sound classification has evolved from applications in speech and music analysis to more advanced uses in safety and security, where it can help detect and respond to emergencies. Early approaches involved using machine learning algorithms on audio data to recognize alarming sounds, such as screams, as part of emergency response systems. These systems, often based on hardware like Raspberry Pi or ARM controllers, aimed to provide real-time notifications in unsafe situations but were limited by bulkiness, making them impractical for personal, everyday use.

With the advent of deep learning, new methods have emerged for classifying sounds more accurately in noisy, dynamic environments. For instance, convolutional neural networks (CNNs) have shown promise in recognizing urban sounds, which could include potential threats like gunshots, breaking glass, or distress calls. Additionally, emotion recognition through speech analysis has been explored for detecting distress in voices, providing indirect indicators of danger. Although effective for identifying emotional states or distinguishing environmental sounds, these systems often struggle with recognizing physical threats in real-world, individual-level scenarios.

Other research focuses on unconventional applications, such as distinguishing drone sounds to monitor urban airspaces or classifying sounds within autonomous vehicles to detect passenger altercations. However, these systems remain limited to specific environments and lack adaptability for broader, real-time personal safety use cases. The challenge lies in creating user-friendly, portable systems that can effectively classify sounds in diverse settings, without relying on bulky hardware, to address the everyday safety needs of individuals.

CHAPTER 3

METHODOLOGY

3.1 PROPOSED APPROACH

3.1.1 SOFTWARE OVERVIEW

The software for this system operates as a mobile application, leveraging the device's built-in microphone to continuously monitor and classify audio signals in real time. It employs machine learning models, including LSTM, 1D CNN, and 2D CNN, trained on a specialized dataset to recognize sounds associated with potential danger, such as screams or gunshots. Upon detecting a suspicious audio pattern, the application immediately triggers an alert, sending the victim's location to emergency contacts and local authorities to ensure timely intervention.

3.1.2 OBJECTIVES

The goal of this system is to provide a real-time audio detection and classification tool that enhances personal safety by identifying and responding to potentially dangerous audio events through a smartphone application. By leveraging the device's built-in microphone, the system aims to detect suspicious sounds, such as screams, gunshots, or glass breaking, with high accuracy, minimizing false detections. Upon detecting such sounds, the application automatically shares the user's geographical location with designated emergency contacts and nearby law enforcement, enabling rapid response and increasing the likelihood of timely intervention. The objectives include achieving robust classification accuracy through the use of machine learning models trained on a diverse dataset of environmental sounds, as well as ensuring that the system is both accessible and effective on standard smartphone hardware without requiring specialized equipment. Through these features, the application aims to empower users by providing a reliable safety tool that works seamlessly in everyday environments.

3.1.3 INPUT AUDIO CLASSIFICATION

The main objective of the proposed system is to detect and classify the victim's live audio signals for immediate rescue. The system is intended to deliver its excellence as an application in any smartphone and it uses the default microphone configuration of it. On detection of suspicious audio patterns from the live input audio from microphone, the geographical location of the victim is shared to the emergency contacts in the phone as well as to the police patrol. The drawbacks inferred from the current violence detection scenarios related to audio event detection and classification accuracy are addressed for effective functioning which plays a vital role in avoiding false event classifications meanwhile ensuring the victim's safety through high classification accuracy. To carry out accurate prediction, training and testing of Kaggle dataset is carried out in 3 machine learning models namely LSTM, 1D CNN and 2D-CNN which is illustrated in figure 2 below:

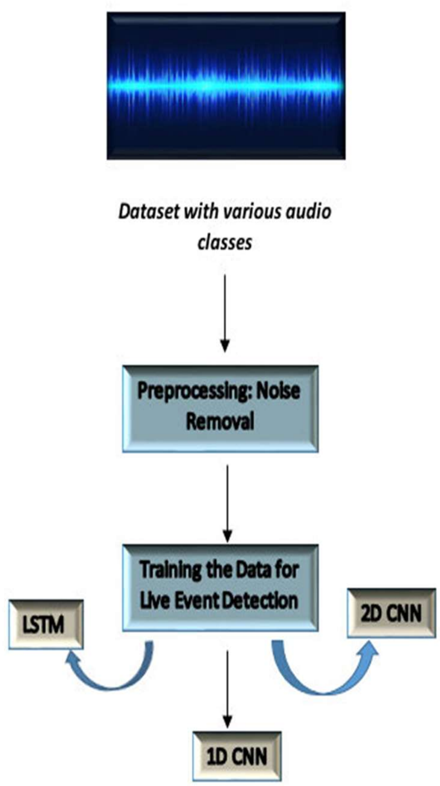


FIGURE 2. **Training the dataset using ML models for live event detection.**

3.1.4 DATASET

The audio dataset used in this work consists of 13 classes (types) of audio signals, namely air conditioner, car horn, children playing, dog bark, drilling, engine idling, fire crack ling, glass breaking, gunshot, jackhammer, scream, siren, and street music. Of these 13 classes, fire crackling, glass breaking, gunshot, and screaming are identified as audio types related to a potentially dangerous environment. The dataset described here is a customized dataset and is built by taking into account and combining parts of data from three different audio datasets in order to meet the requirements of the problem statement under research. The three datasets mentioned here are the UrbanSound8K dataset, which is a dataset with audio data for 50 different environmental sounds, and which is a dataset for the 'screaming' noise as this sound type is a crucial one for the requirement of the project, which focuses on the detection of threat from the ambient noise. Although the three original datasets from which the dataset for this project has been built, have over 50 different classes of audio signals, only 13 are kept for the final analysis as it would take less computational time for only 13 classes to be trained on the three deep learning models built during the course of this project, as compared to the larger size of a dataset with over 50 classes, taking into account the fact that this research focuses more on the successful development of a working prototype than a production-level software.

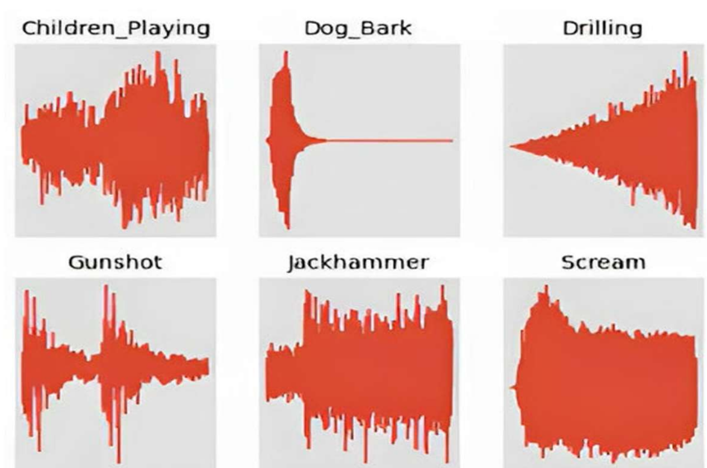


FIGURE 3. The audio signals in the time domain for one sample audio each from 6 of the 13 classes considered in the research.

3.1.5 EXPLORATORY DATA ANALYTICS (EDA)

Exploratory Data Analysis (EDA) in audio signal processing is a crucial step in understanding and preparing audio data before applying machine learning or deep learning models. By examining the data, EDA helps identify patterns, detect anomalies, and ensure data quality, which improves the accuracy and performance of predictive models. Key stages in EDA for audio data include sampling, where a subset of data is taken to represent the whole; converting time-domain signals to frequency domain, which reveals the sound's frequency components creating Mel spectrograms, which capture the audio's time-frequency information in a format that resembles human auditory perception; and cleaning the sound data to remove noise or irrelevant information. These steps facilitate deeper insights into the structure and characteristics of the audio signals, optimizing the data for further analysis and model training.

3.1.5.1 SAMPLING

Sampling in audio signal processing is the method of converting a continuous audio signal into a digital signal by measuring the signal's amplitude at evenly spaced intervals, or sampling points. This process allows the analog audio wave to be stored and processed in a digital form, making it suitable for applications such as digital playback, editing, and transmission.

The rate at which samples are taken, known as the sampling rate, significantly impacts the quality and accuracy of the digital representation. Higher sampling rates, such as 44.1 kHz (used in audio CDs) and 48 kHz (standard in professional audio), capture more detail from the original signal. The Nyquist Theorem establishes that to accurately reproduce the audio without distortion or loss of information, the sampling rate must be at least twice the highest frequency in the audio signal.

Common sampling rate: 44.1KHz ,48 kHz

Nyquist Theorem: $fs \geq 2\ fmax$

fs :sampling rate, fmax : maximum frequency component in the signal

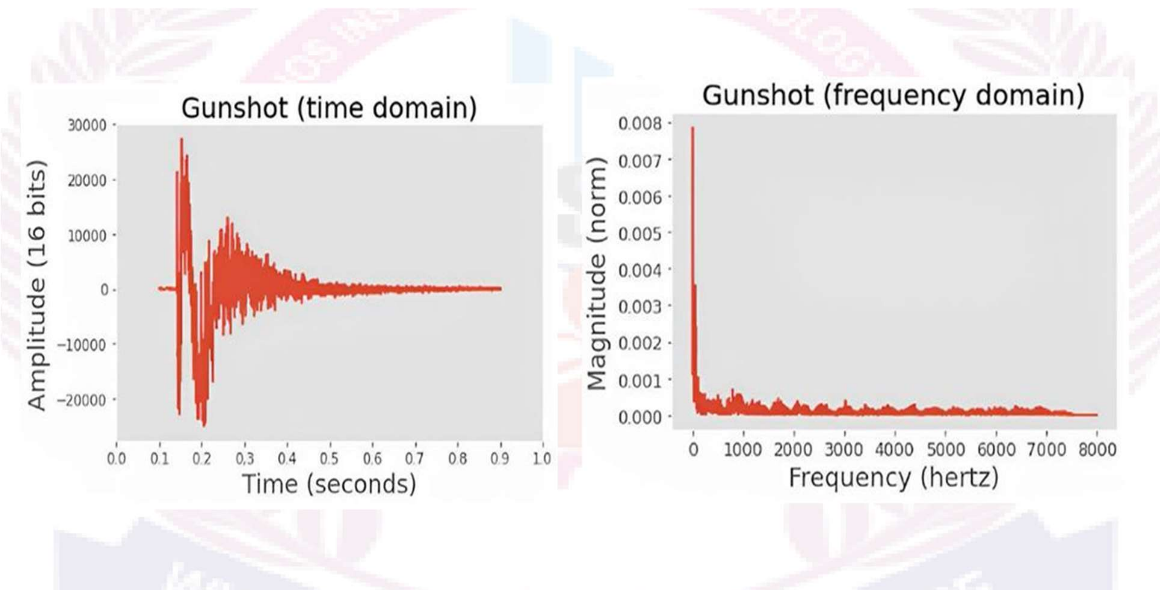
3.1.5.2 CONVERTING TIME DOMAIN INTO FREQUENCY DOMAIN

THE FAST FOURIER TRANSFORM (FFT)

It is a method used to convert an entire time-domain signal into the frequency domain, providing a global representation of the signal's frequency content. It is ideal for stationary or periodic signals where the frequency characteristics do not change over time. The FFT is efficient in analyzing signals that remain constant, allowing for a detailed understanding of the frequency spectrum. However, it has a limitation in that it cannot provide time-specific information, making it unsuitable for analyzing non-periodic or transient signals.

SHORT-TIME FOURIER TRANSFORM (STFT)

It is designed to handle non-stationary signals by dividing the signal into short segments and converting each segment into both time and frequency domains. This method provides a time-frequency representation, which is particularly useful for analyzing signals that change over time, such as speech or music. While the STFT allows for the observation of how frequency components evolve in time, its main limitation is that the time resolution is dependent on the window size used for segmentation, which can compromise the accuracy of frequency det



3.1.5.3 MEL SPECTROGRAM REPRESENTATION

MEL SPECTROGRAM

A Mel Spectrogram is a time-frequency representation of an audio signal that has been transformed using the Mel scale, which is designed to match the way humans perceive pitch. The Mel scale is nonlinear, compressing higher frequencies and expanding lower frequencies to reflect the human ear's sensitivity to different pitches. However, as the frequency increases, the changes on the Mel scale become less pronounced, with higher frequencies being compressed more. This characteristic allows the Mel spectrogram to reflect how humans perceive sound, giving more importance to lower frequencies and capturing the nuances that are most perceptible to the human ear. To generate a Mel spectrogram, the audio signal is passed through a Mel filterbank, is a series of filters applied to the frequency components of an audio signal to approximate how humans perceive sound frequencies. These filters emphasize lower frequencies by using the Mel scale, creating a compressed representation where each filter output corresponds to a specific frequency band, enhancing the most perceptible aspects of the signal.

MEL SCALE

The Mel scale is a perceptual scale of pitch that reflects how humans perceive changes in frequency. Unlike a linear frequency scale, the Mel scale compresses higher frequencies and expands lower frequencies, as the human ear is more sensitive to changes in pitch at lower frequencies than at higher ones. This non-linear scale is designed to model human auditory perception more accurately, making it particularly useful for tasks such as speech processing, where the focus is on features relevant to human hearing. The Mel scale is commonly used in audio signal processing, especially in the creation of Mel spectrograms, which are widely applied in areas like speech recognition and music analysis.

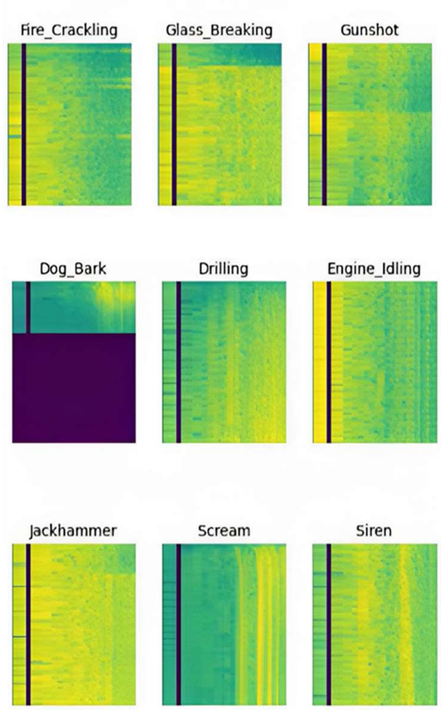


FIGURE 6. The decibel spectrogram for one sample audio each from 12 of the 13 classes considered in the research (the same as considered in the time domain plots above).

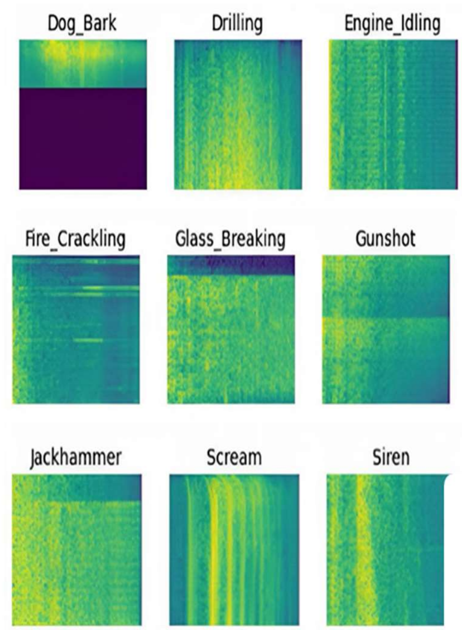


FIGURE 7. The Mel spectrogram for one sample audio each from 12 of the 13 classes considered in the research.

3.1.5.3 CLEANING THE SOUND DATA

To prepare audio data for analysis, each file is organized into class-specific folders and converted to the ".wav" format for faster loading, despite its high memory usage. To standardize diverse real-world sounds, audio files are converted to mono and down-sampled to 16,000 Hz, reducing computation time during training and ensuring uniformity across data. Silent regions within the audio, which add minimal value, are removed using a custom signal envelope that filters out low-magnitude sections, retaining only relevant data for analysis. The cleaned audio is split into 1-second intervals, or "delta time" segments, to facilitate efficient model training on manageable samples. This cleaned, segmented data is then stored in a new directory, organized by class, to streamline model training on consistent, high-quality input.

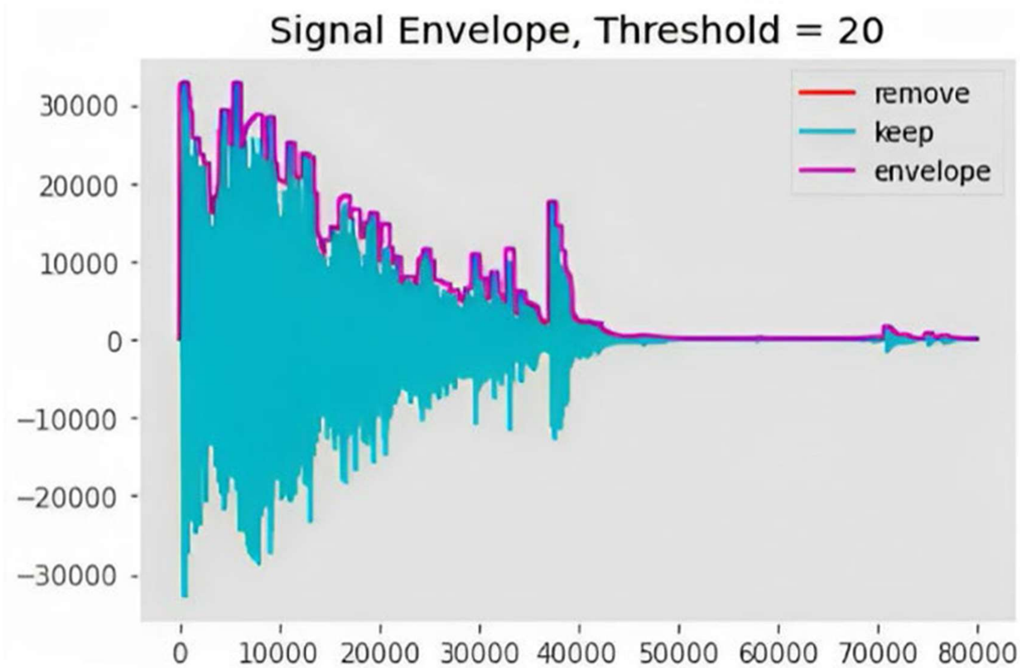


FIGURE 8. The signal envelope for one of the audio signals corresponding to "glass breaking". Here, since the threshold magnitude is considered as 20, any part of the signal below that threshold is not considered, and this can be verified with the help of the non-silent signal envelope created around the signal (visible in purple).

3.1.6 APPLIED DEEP LEARNING MODELS

This project explores three deep learning models—LSTM, 1D-CNN, and 2D-CNN—each offering unique advantages for training, analyzing, and classifying audio data. The prediction module is designed to allow selection among these models based on the specific needs of the audio input. A custom data generator prepares and loads cleaned audio data into the models in efficient batches, enabling processing across multiple GPUs for optimal performance during training.

3.1.6.1 THE 1D-CNN (1-DIMENSIONAL CONVOLUTIONAL NEURAL NETWORK) MODEL

The 1D-CNN model is designed to capture significant features within fixed-length segments of data, making it effective for time-series data such as audio signals. This model wraps 1D convolutions over time with time-distributed layers, allowing it to process the frequency spectrum of audio signals across time. A permute layer reorders the input data dimensions to align with the time-distributed format, setting time as the primary dimension for the network. The model then applies convolution layers with different activation functions, such as hyperbolic tangent and ReLU, enabling the network to capture nonlinear patterns and avoid issues like vanishing gradients. As it progresses through layers, the network gradually expands its feature space, building complexity until it reaches a dense representation that can effectively feed into classification layers.

Towards the end, a GlobalMaxPooling layer captures the last dimension of features, ensuring efficient use of the model's resources for training. To prevent overfitting, the model incorporates dropout and regularization layers. The final classification is achieved through a dense layer that includes an output SoftMax activation function, converting the outputs into probability vectors for each class, thus enabling categorical classification. By using categorical encoding, the model effectively predicts the class of audio data based on the probability calculated across possible outcomes.

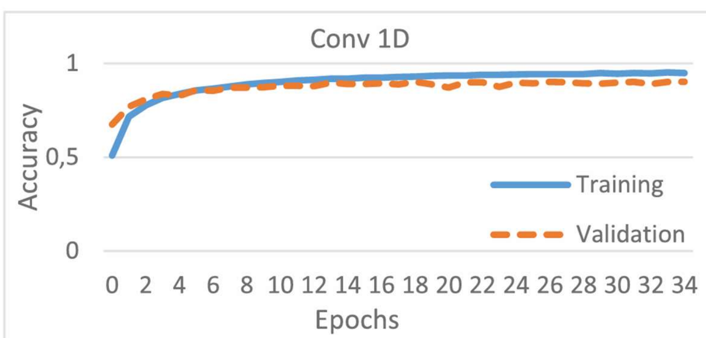


FIGURE 9. On running for 35 epochs, the maximum accuracy reached for the 1D-CNN model is found to be 95.2%, with a maximum validation accuracy of 90.2%. The metric used here for considering the best results is the validation loss. The Training (train) vs Validation (test) can be seen in this figure.

3.1.6.2 THE 2D-CNN (2-DIMENSIONAL CONVOLUTIONAL NEURAL NETWORK) MODEL

The 2D-CNN model, rooted in the Lenet-5 architecture, is highly effective for analyzing data with spatial dimensions, such as audio spectrograms, due to its ability to capture spatial relationships within the input. This is achieved by using Conv2D layers, where the kernel moves across two dimensions. As the model processes the spectrogram, it recognizes patterns by evaluating adjacent features, which are key in learning the frequency patterns within audio data. The model's architecture consists of successive Conv2D layers, each with progressively more activation units (e.g., 8, 16, 32), and uses ReLU activation functions, except in the first layer, which uses a hyperbolic tangent activation function for improved performance in speech and language-related tasks. MaxPooling layers are added between convolutional layers to downscale feature maps, which helps to retain the most relevant information from each region.

The final layers of the 2D-CNN model closely resemble those in a 1D-CNN, with flattening, dropout, and regularization layers to prevent overfitting. After the last Conv2D layer, the model applies a flattening layer that reduces the output to a one-dimensional array for further processing, avoiding excessive memory usage. This step is particularly crucial because the time intervals set during preprocessing (e.g., 1-second intervals) influence the final dimensions of the input, which affects memory requirements and computational efficiency. The final classifier comprises two dense layers, using 64 activation units in the first dense layer and applying a SoftMax activation function in the last layer, which outputs prediction probabilities for all class.

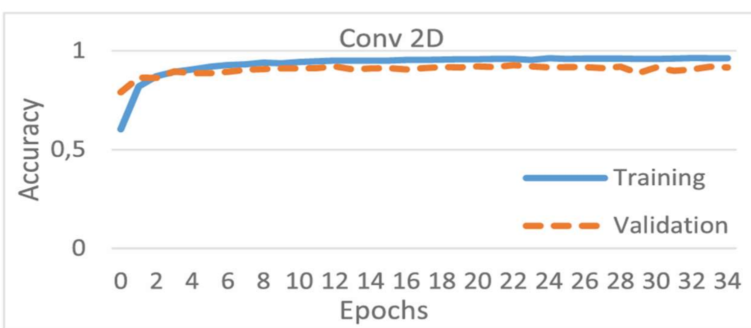


FIGURE 10. On running for 35 epochs, the maximum accuracy reached for the 2D-CNN model is found to be 96.3%, with a maximum validation accuracy of 92.7%. The metric used here for considering the best results is the validation loss. The Training (train) vs Validation (test) can be seen in this figure.

3.1.6.3 THE LSTM (LONG SHORT-TERM MEMORY) MODEL

LSTM (Long Short-Term Memory) networks, a type of Recurrent Neural Network (RNN), are particularly well-suited for tasks involving sequences of data, such as audio, video, and speech recognition. Unlike traditional feedforward neural networks, LSTMs have feedback connections that allow them to process entire data sequences, capturing long-term dependencies. An essential component of LSTM is the "cell state," which maintains information over time, with gates that regulate the flow of data. These gates, controlled by sigmoid layers, determine which information should be added to or removed from the cell state. The LSTM model in this research is designed to learn features over time by processing input in batches, using a time-distributed dense layer and a Bidirectional LSTM layer to process data in both forward and backward directions for more accurate gradient updates.

The LSTM network progresses by applying a Skip-Connection technique, which concatenates the features from the time-distributed dense layer with those from the Bidirectional LSTM layer. This combined feature set is then processed through additional dense layers with activation units, followed by MaxPooling and Flattening layers. A 1D-MaxPooling layer is used due to the loss of channel information during reshaping. The final classification process involves dropout and regularization layers to minimize overfitting, with two dense layers (one with 32 units and the other with sound classes) used for classification. The output is generated using a SoftMax activation function, which provides a probability distribution for each class, similar to the classification strategies used in the 1D-CNN and 2D-CNN models.

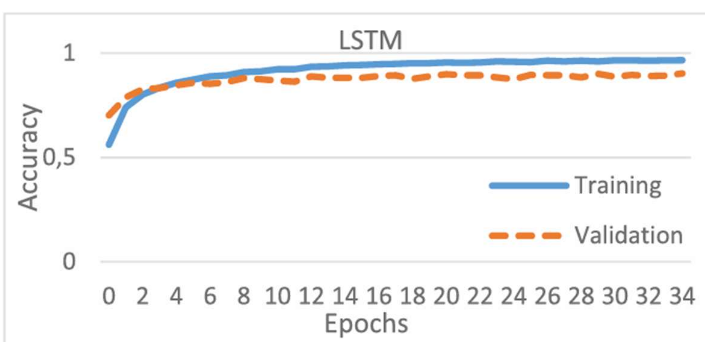


FIGURE 11. On running for 35 epochs, the maximum accuracy reached for the LSTM model is found to be 96.6%, with a maximum validation accuracy of 90.3%. The metric used here for considering the best results is the validation loss. The Training (train) vs Validation (test) can be seen in this figure.

3.1.7 THE LIVE AUDIO RECORDING AND PREDICTION MODULE

The prediction module processes real-world audio data by down-sampling the recorded audio to a mono channel with a 16,000 Hz sampling rate and creating a non-silent signal envelope to focus on relevant parts of the signal. The audio is then batched into 1-second intervals, and predictions are made using the argmax function on hot-encoded probabilities, averaging them over the entire sample for final classification. If the audio is identified as related to a potentially hostile environment, such as fire crackling, glass breaking, gunshots, or screams, an automatic alert is sent via email, SMS, and WhatsApp to registered contacts. The email includes the recorded audio file, allowing recipients to listen to the event that triggered the alert, with the recording saved and processed by the audio recording module.

The audio recording module captures live audio from the environment and stops recording when no noise is detected. It then performs several pre-processing steps, including trimming silence from both ends of the recording, normalizing the volume across the entire recording, and adding a 0.5-second audio padding at both ends to ensure no crucial data is lost during playback. Once pre-processing is complete, the audio is saved as a ".wav" file at a specified location. The prediction module is then automatically triggered to analyze the recorded audio using one of the three deep learning models defined by the user, performing all necessary steps to identify the class of the audio.

CHAPTER 4

EVALUATION

This research aimed to identify real-world audio data using deep learning-based multi-class classification and determine if the identified sound type indicates a potentially dangerous or threatening environment. The focus was to create a threat alert system that automatically sends an alert via email, SMS, and WhatsApp, including an audio recording of the environment. The system successfully records, preprocesses audio, runs the prediction model, and issues an immediate alert when a threat is detected.

4.1 PERFORMANCE OF DEEP LEARNING MODELS

4.1.1 CONFUSION MATRIX

The confusion matrix is a crucial tool in the performance analysis of deep learning models, providing a detailed breakdown of how well the model performs in classifying data. It displays the true positives, true negatives, false positives, and false negatives for each class, helping to identify where the model makes correct predictions and where it misclassifies data. In the context of multi-class classification tasks, the confusion matrix allows for a clearer understanding of the model's accuracy by showing which classes are more likely to be confused with each other. This information is vital for evaluating model performance beyond just accuracy, as it highlights specific areas where improvements, such as better feature extraction or model tuning, may be needed. It includes four key components: True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN), which indicate the model's correct and incorrect predictions. For binary classification, the matrix is a 2x2 table, with rows representing actual classes and columns representing predicted classes. These components are essential for calculating metrics like accuracy, precision, recall, and F1-score to assess the model's performance.

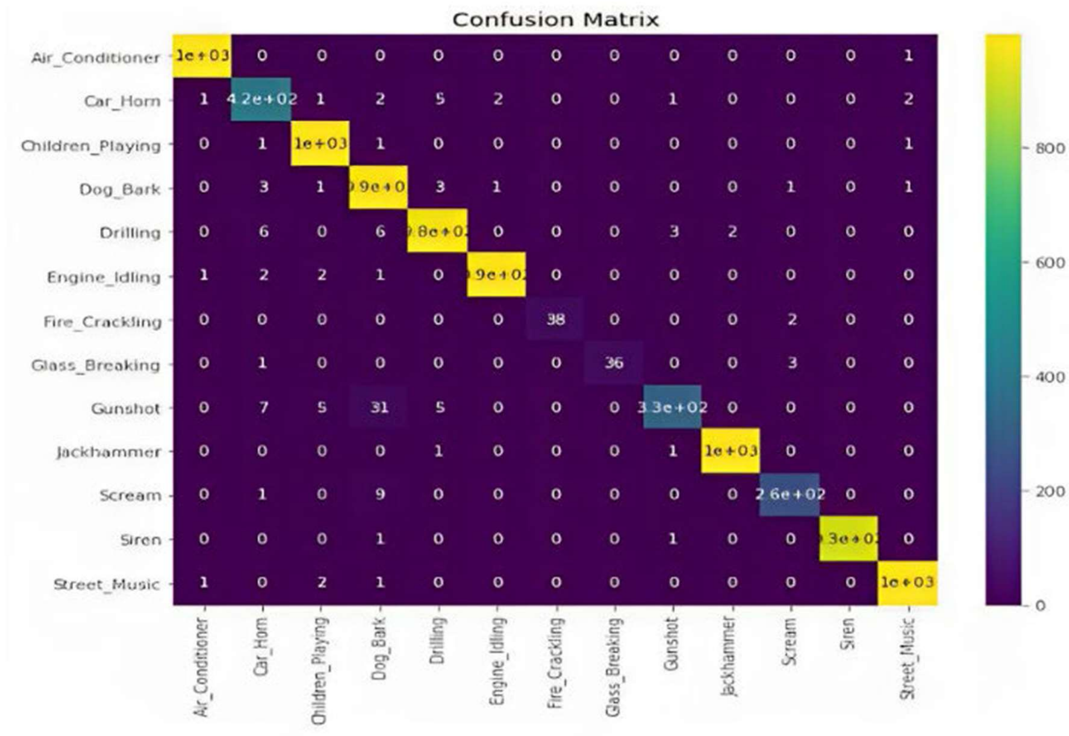


FIGURE 12. The confusion matrix attained in live event prediction.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1\text{-}score = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

CHAPTER 5

CONCLUSION

A new software-based threat identification system has been developed to detect dangerous situations from an individual's ambient noise and automatically inform their emergency contacts. The system uses 1D-CNN, 2D-CNN, and LSTM models to record live audio, identify the sound type, and predict the threatful situation, achieving accuracies of 95.2%, 96.3%, and 96.6%, respectively, with an average accuracy of 96.03%. The system offers users the option to choose between models, with the choice becoming more useful as the dataset and live audio complexity increase. It effectively combines deep learning with audio signal processing to create a reliable system for real-time threat detection. Furthermore, the research highlights the potential for future improvements, including expanding the dataset and refining the models for more complex situations. This system offers a promising approach to enhancing safety and security by leveraging advanced machine learning techniques to detect and respond to potentially harmful environments. Ultimately, the research successfully addresses the problem of threat detection and alert systems, leveraging deep learning and audio analysis to detect live events from ambient sounds.

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